CS325 Implementation Assignment 3

PROBLEM 1:

Given a set of points (x1, y1), . . ., (xi, yi) find a line y = ax + b that minimized the maximum abosolute deviation from all points. Min(max|axi + b – yi|

Variables:

a, b, z (max absolute deviation)

Objective:

Minimize z

Constraints:

For each point I = 1. . . , n

Introduce z to replace the max by requiring error at every point to be at most z. We also have to account for absolution value so

Constraint 1: axi + b – yi <= z

Or

axi + b – z <= yi

Constraint 2: -(axi + b – yi) <= z

Or

-axi - b – z <= -yi

Constraint 3:

z >= 0

Optimal solution:

Slope of best fit: a = 1.7143

y-int: b = 1.8571

max absolute deviation from any point: E = 0.5714

Output:

	pcost	dcost	gap	pres	dres	k/t
0:	0.0000e+00	-9.7145e-17	2e+00	1e-01	3e-17	1e+00
1:	1.6598e-01	1.3586e-01	5e-01	4e-02	4e-16	3e-01

2: 5.6282e-01 4.8291e-01 3e-01 2e-02 3e-15 4e-02

3: 5.6736e-01 5.6299e-01 1e-02 9e-04 3e-16 2e-03

4: 5.7139e-01 5.7134e-01 2e-04 9e-06 1e-15 2e-05

5: 5.7143e-01 5.7143e-01 2e-06 9e-08 9e-16 2e-07

6: 5.7143e-01 5.7143e-01 2e-08 9e-10 3e-16 2e-09

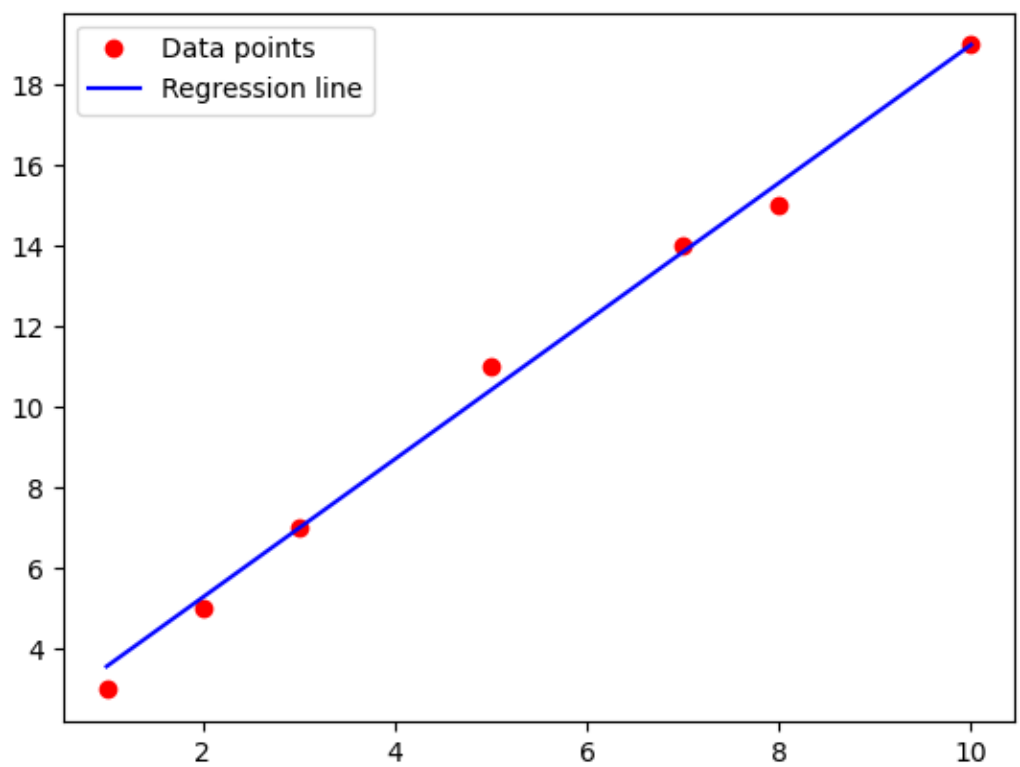
Optimal solution found.

Optimal a: 1.714285713007345

Optimal b: 1.8571428630166555

Optimal t (max absolute deviation): 0.5714285673244874

Plot saved as regression_plot.png:



Problem 2:

Daily average temp in Corvallis modeled using a linear trend plus two sinusoidal components.

$T(di) = x0 + x1*di + x2*\cos(2\pi di/365.25) + x3*\sin(2\pi di/365.25) + x4*\cos(2\pi di/3908.18) + x5*\sin(2\pi di/3908.18)$

Variables: =

x0, x1, x2, x3, x4, x5, z

Objective: minimize z

Constraints: for each data point i

Constraint 1: x0 + x1*di + x2*cos(...) + x3*sin(...) + x4*cos(...) + x5*sin(...) - z <= Ti

Constraint 2: -x0 - x1*di - x2*cos(...) - x3*sin(...) - x4*cos(...) - x5*sin(...) - z <= -Ti

Constraint 3: z >= 0

Optimal Values

x0 = 8.0214

x1 = 0.00010695 (daily drift in °C)

x2 = 4.2809

x3 = 8.18686

x4 = -0.7906

x5 = -0.29536

Minimum maximum absolute deviation (E) = 14.23554

Output:

	pcost	dcost	gap	pres	dres	k/t
0:	-7.0605e-21	-0.0000e+00	2e+01	2e+00	5e-15	1e+00
1:	2.3167e+00	1.4641e+00	2e+01	1e+00	2e-12	1e-02
2:	4.4763e+00	3.8533e+00	1e+01	1e+00	8e-13	5e-03
3:	4.9624e+00	4.3398e+00	1e+01	1e+00	9e-13	5e-03
4:	8.0863e+00	7.6275e+00	1e+01	8e-01	3e-12	2e-03
5:	9.4040e+00	9.0987e+00	7e+00	5e-01	3e-13	1e-03
6:	9.5334e+00	9.2309e+00	7e+00	5e-01	5e-12	1e-03
7:	1.0041e+01	9.7627e+00	6e+00	5e-01	3e-12	8e-04
8:	1.2147e+01	1.2006e+01	3e+00	2e-01	1e-11	3e-04
9:	1.3242e+01	1.3181e+01	1e+00	1e-01	3e-12	1e-04
10:	1.3443e+01	1.3402e+01	9e-01	7e-02	4e-12	4e-05
11:	1.4020e+01	1.4009e+01	2e-01	2e-02	3e-11	1e-05
12:	1.4183e+01	1.4181e+01	6e-02	4e-03	1e-10	2e-06
13:	1.4224e+01	1.4223e+01	1e-02	9e-04	4e-11	4e-07
14:	1.4235e+01	1.4235e+01	1e-04	1e-05	4e-10	4e-09
15:	1.4236e+01	1.4236e+01	1e-06	1e-07	1e-10	4e-11
16:	1.4236e+01	1.4236e+01	1e-08	1e-09	1e-10	4e-13

Optimal solution found.

Optimal solution found:

x0 = 8.021419720760887

x1 = 0.00010694836297869377 (daily drift in °C)

x2 = 4.2808907471513935

x3 = 8.186857819322274

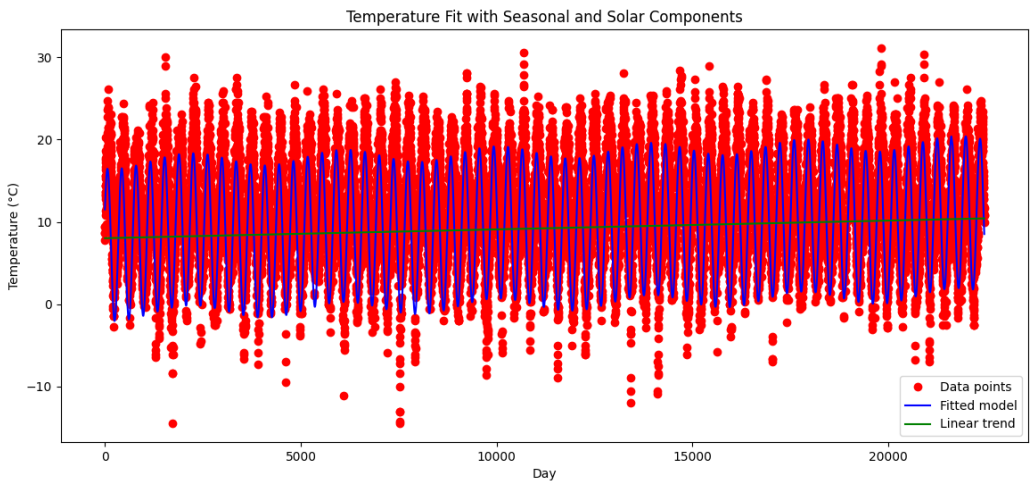
x4 = -0.7906307941596648

x5 = -0.29536020812959285

Minimum maximum absolute deviation (E) = 14.235540281788637

Estimated annual drift (x1 * 365.25) = 0.0390628895779679 °C/year

Plot saved as 'temperature_fit.png':



0.00010694836297869377 * 365.25 * 100 = 3.9062889578

3.906 Degrees C / Century

Since x1 is positive, Corvallis is experiencing a warming trend of 3.906 degrees C per century.